

CURRICULUM VITAE DR. EVA C. HERBST

GROUP LEADER/OBERASSISTENTIN IN MUSCULOSKELETAL BIOMECHANICS OF THE SHOULDER

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LANGUAGES: Native German and English speaker

LINKS: [Website](#) - [Github](#) - [GoogleScholar](#) - [Figshare](#) - [Orcid](#)

EDUCATION

OCT 2016 - OCT 2019	PhD in Biomechanics and Palaeontology Thesis: "Computational Analysis of the Evolution of Amphibian Locomotor Modes" Advisor: John Hutchinson, Structure & Motion Lab, Royal Veterinary College, London *defense 15 July 2020, certificate 2021 due to Covid
AUG 2012 - MAY 2016	B.A. in Integrative Biology U.C. Berkeley, Distinction in General Scholarship (equivalent of cum laude)
OCT 2013 - JUN 2014	Certificate of Higher Education in Biomedical Sciences Durham University, Year of Study Abroad

EMPLOYMENT & RESEARCH EXPERIENCE

MAR 2023 - PRESENT	Group Leader/Oberassistentin Computational Shoulder Biomechanics ETH & Schulthess Clinic, Zurich Specializing in dynamically coupled finite element and multibody dynamic analyses, development of patient specific modeling pipeline. <i>Postdoctoral researcher March 2023 - May 2026, promoted to Oberassistentin/Group Leader June 1st, 2026</i>
DEC 2019 - NOV 2022	Postdoctoral Researcher: Skull Biomechanics of Triassic Reptiles Palaeontological Institute & Museum, University of Zurich Anatomical reconstruction of fossil skulls from imaging data, finite element analyses to investigate function
OCT 2019 - DEC 2019	OATech+ Network Early Career Researcher Placement: Knee Osteoarthritis Skeletal Biology Group, Royal Veterinary College London Development of segmentation method for murine trabecular bone, analysis of relationship between trabecular bone and osteoarthritis
MAY 2013 - JUL 2016	Undergraduate Research Assistant: Anatomy and Biomechanics Projects U.C. Berkeley & University of Missouri Reconstruction of fossil skull anatomy, finite element analysis

TECHNICAL SKILLS

- biomechanical model development:** development of patient specific, muscle-driven finite element shoulder models, virtual joint range of motion analysis
- medical imaging:** 3D rigid and deformable registration, landmark transfer, segmentation method development (AI model training and non-AI methods), image anonymisation
- in vivo kinematic analysis:** biplanar fluoroscopy
- ex vivo kinematic analysis:** cadaveric analysis with optical motion capture (Qualisys, Optitrack)

- **programs and languages:** Artisynth, Blender, 3D Slicer, Avizo, Mimics, Abaqus, Maya, Python, Java, R, MonaiLabel, XNAT, RedCap
- **additional skills:** dissections, photogrammetry, freeform 3D modeling

HONORS & AWARDS

2021	D. Dwight Davis Award, Society of Integrative and Comparative Morphology Best student oral presentation in the Division of Vertebrate Morphology
2020	Swiss Commission of Palaeontology Prize Best presentation in palaeontology given at the Swiss Geoscience Meeting
2016	Franklin M. Henry Award, Integrative Biology, UC Berkeley Outstanding achievement in human performance and health research
2013, 2015	Dean's Honors, UC Berkeley For exceptional grade point average

GRANTS & FUNDING

2025/26	Arthrex Research Grant Implant testing study. Role: PI, grant application, planning testing protocol, conducting implant simulation work, coordinating international research team, legal aspects, and funding	103,356 CHF
2025	ETH Focus Grant Grant for 1 month funding and coaching to develop career starting grant application, awarded based on merit of proposed research project and suitability of candidate	9,745 CHF
2024	ETH Career Seed Award "Age-based Bone Density of the Glenoid: Method Development & Quantification of Patient Variation", Role: PI, project conceptualization, application, supervision, funds for 3 months research assistant	30,000 CHF
2024	Gesellschaft für Arthroskopie und Gelenkchirurgie Grant Grant for establishing a clinical imaging database of shoulder instability Patients, Role: grant application, project planning and implementation	10,000 EUR
2024	Hamlyn Symposium on Medical Robotics Workshop Funding Organisation of Workshop: Open-Source Software for Surgical Technologies, London, June 2024	2,500 GBP
2023	Digital Switzerland Boost Programme Grant for attending Advanced 3D Slicer Programming Course	420 CHF
2021	ImagingBioPro Network Online Educational Material Grant Development of educational materials and code: mesh manipulation and trabecular segmentation	1,000 GBP
2020	University of Zurich GRC Grant Organization of finite element analysis conference (> 200 participants), resource platform development	10,000 CHF
2019	OATech+ Network Biomechanics and Mechanobiology Pump Priming Grant Research project: 3D trabecular architecture as a biomarker to identify and monitor knee osteoarthritis (role: grant application, project coordination and execution)	10,000 GBP
2019	OATech+ Network Early Career Researcher Placement Grant Placement with Prof Andrew Pitsillides at RVC to work on osteoarthritis project (see above)	3,000 GBP
2018 2019	Royal Veterinary College Foreign Travel Fund 300 GBP each year to present research at ICVM and SICB conferences	600 GBP
2016	Research Experience for Undergraduates, National Science Foundation	3,500 USD

PUBLICATIONS & PRESENTATIONS

- 20 published papers (10 first author), 36 conference presentations (20 presenting author), details [here](#)
- 11 invited talks and workshops, details [here](#)

TEACHING

- **Lecture for MSc Course “Clinical Challenges in Musculoskeletal Disorders”:** *Current Research in Shoulder Biomechanics*. ETH. 2024, 2025, 2026
- **Lectures MSc Courses Bio 262 & Bio 267:** *Using Computer Tools to Investigate Biomechanics of Animals*. University of Zurich. 2020-2022.
- **Lab MSc Bio 262: Evolutionary Morphology of Vertebrates: Issues and Methods.** Role: leading and designing anatomy practicals. University of Zurich. 2021, 2022
- **Lab MSc Bio 267: Paleobiology and Evolution of Vertebrates.** Role: leading and designing anatomy practicals. University of Zurich. 2021, 2022
- **Comparative BSc Animal Locomotion Course.** Role: leading research paper seminar. Royal Veterinary College, London. 2017, 2018.

SUPERVISION

1. | **Max Modelhart**, *Medical PhD on glenohumeral joint morphology and density analysis (official title to be determined)*, Paracelsus Medical University (PMU), Salzburg, Austria (Co-supervisor, Ongoing)
2. | **Riccardo de Luca**, *Medical student research internship project on statistical shape modeling of juvenile shoulder joints*, internship at Schulthess Clinic (supervisor, Ongoing)
3. | **Maartje van Beurden**, Master's Research Internship: *Effect of posture on rTSA range of motion*, ETH Zurich (2026)
4. | **Lucy McCarthy**, Master's Semester Project: *Distal Humerus Reconstruction from Routine Clinical CT Scans using Statistic Shape Modelling and Deformable Registration*, ETH Zurich (2026)
5. | **Anna Luong**, Master's Thesis: *Development and analysis of personalized glenohumeral joint contact mechanics in healthy subjects*, ETH Zurich (2025)
6. | **Liam Roth**, Master's Semester Project: *Cartilage morphology effects in patient specific biomechanical finite element glenohumeral joint models*, ETH Zurich (2024)
7. | **Jan Heres**, Master's Thesis: *Reconstruction of a patient-specific model of the humerus bone*. University of West Bohemia (acting as external supervisor) (2024)
8. | **Dennis Agbanyim**, Master's Thesis: *Phantomless bone density calculation: developing research software for patient-specific shoulder modeling*, ETH Zurich (2024)
9. | **Flavia Stettler**, Semester Project: *In vivo glenohumeral translations during movements in healthy subjects*, ETH Zurich (2024)
10. | **Kehan Pan**, Semester Project: *Skull finite element analysis of Askeptosaurus*, ETH Zurich (2022)

OPEN SOURCE WORK

METHODS & CODE

- Python-based [Blender plugin](#) for modelling 3D muscles
- Method for [visualizing joint range of motion](#)
- Method for [trabecular bone segmentation](#), plugin ft. on [official Avizo website](#)

	• Blender remeshing guide for FEA
FEZ INITIATIVE	Finite element resource development and dissemination: Finite Element Zurich
CT DATA & 3D MODELS	Available on Figshare

PROFESSIONAL SERVICE, LEADERSHIP, & OUTREACH

2026	Session Chair (Scapulothoracic Kinematics) and Jury Member Best Poster Presentation Award, International Shoulder Group Conference
2025	Track Chair (Shoulder Track) and Reviewer, European Society of Biomechanics
2025	Biomechanics Discovery Outreach Event for Children, European Society of Biomechanics
2024	Organized Workshop “Open-Source Software for Surgical Technologies” at Hamlyn Symposium for Medical Robotics
2020	Organized Finite Element Analysis Conference and Workshop (200+ participants)
2018	Session Chair, Society of Integrative and Comparative Biology Annual Meeting, San Francisco.
PEER REVIEW	PNAS, Clinical Biomechanics, Journal of Biomechanics, The Anatomical Record, Journal of Anatomy, Integrative Organismal Biology, Methods in Ecology and Evolution, Integrative and Comparative Biology, Canadian Journal of Earth Sciences

PROFESSIONAL DEVELOPMENT & CERTIFICATES

ONGOING	Foundations of Teaching and Learning , ETH Zurich
2023	Advanced 3D Slicer Course: Scripting and Customization, Kitware
2022	Good Clinical Practice Module 1 and certification
2022	Data Analysis for Medical Research using R, UZH
2021	Scientific Programming with Python , Physics Department, UZH
2020	Open Life Science Course
2020	SlicerMorph 3D Morphometrics Course
2019	Avizo Course, 3DMAGINATION Ltd.
2018	MatLab Fundamentals Course
2017	Teaching and Learning in Higher Education Certificate, Royal Veterinary College, London